

DCIT 204/308: DATA STRUCTURES AND ALGORITHMS I & II

JOINT SEMESTER PROJECT

MEETING 2 AGENDA AND PROJECT KICK-OFF PLAN

Group Size: 16 Members

Provisional Project Domain: Courier/Food Delivery Service

Proposed Project Title: *Legon Smart Food Delivery and Rider Dispatch Optimizer*

Meeting Date: 28/07/2026

Meeting Time: 8:00 pm

Venue/Platform: Online

Meeting Chairperson: Adom

Minutes Secretary: Solomon

1. Background

The group held its first meeting to review the semester project brief, identify members' skills and discuss possible project domains.

Only 6 out of the 16 members attended the first meeting. A skills poll and project-domain poll were later shared to gather additional information from the remaining members.

Based on the domain poll:

Proposed Domain	Votes
Courier/Food Delivery Service	6
Library or Records Management	2
University Campus Service Operations	0

The Courier/Food Delivery Service is currently the preferred option. However, the final decision must be officially approved during Meeting 2.

The project requires a Ghanaian context, custom data structures, algorithms, database integration, testing, performance experiments, documentation and individual accountability.

2. Purpose of Meeting 2

Meeting 2 will serve as the official project kick-off meeting.

The meeting is expected to:

1. Confirm the participation and skills of all 16 members.
 2. Finalise the project domain and title.
 3. Define the exact system scope and operations.
 4. Approve the project leadership and team structure.
 5. Assign development, documentation and presentation responsibilities.
 6. Assign every member one data structure and one algorithm to study.
 7. Agree on the database, development tools and collaboration process.
 8. Assign the first project tasks and deadlines.
 9. Ensure every member understands what is expected before the next progress meeting.
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3. Requirements Before Meeting 2

Every member is expected to complete the following before the meeting:

Requirement	Expected Action
Attendance	Attend the meeting and remain until the meeting ends
Personal details	Be ready to provide full name and index number
Skills confirmation	State whether selected skills are strong, intermediate or areas the member is willing to learn
Role preference	Select a preferred primary role and support role
Project review	Read the project brief or the group summary
Domain decision	Be ready to approve or reasonably object to the proposed food delivery domain
Presentation topic	Be ready to accept one data structure and one algorithm

Availability State the days and times available for group work

Commitment Be prepared to accept a task and deadline

Members who did not complete the poll will be required to provide their skills and preferences during the meeting.

All attendance must be properly recorded because the project brief requires members to sign in and out during meetings.

4. Official Meeting Agenda

Agenda 1: Opening and Attendance

Estimated time: 5 minutes

Activities:

- Welcome members.
- Record the names and index numbers of all members present.
- Record members who are absent.
- Confirm the chairperson and minutes secretary.
- Confirm the official names behind unclear poll display names.

Expected Outcome

A complete attendance record for all 16 members.

Agenda 2: Review of Meeting 1 and Poll Results

Estimated time: 10 minutes

The group will receive a brief summary of:

- Decisions made during Meeting 1.
- Matters that could not be completed.
- Skills-poll results.
- Project-domain poll results.
- Areas that received no volunteers.
- Members who did not complete the polls.

Main Poll Observations

The poll showed interest in:

- Java fundamentals.
- Object-Oriented Programming.
- Data Structures and Algorithms.
- Searching and sorting.
- SQL and database management.
- Testing and debugging.
- Documentation.
- Video recording and editing.
- Research.
- Members willing to learn and support.

The poll also showed that several important technical areas received no direct volunteers, including:

- Trees, heaps and hash tables.
- Graph algorithms.
- Greedy algorithms.
- Dynamic programming.
- Code integration.
- Performance testing.
- Console interface development.
- Git and GitHub.

These areas must still be assigned because they form part of the project requirements.

Expected Outcome

All members understand the current position of the group.

Agenda 3: Final Skills Confirmation

Estimated time: 15 minutes

Each member will briefly state:

1. Full name.
2. Index number.
3. Primary skill.
4. Secondary skill.
5. Skill level:
 - Strong.
 - Intermediate.
 - Beginner or willing to learn.
6. Preferred project responsibility.

7. Weekly availability.

Members may select more than one area, but they must clearly state the area they can lead and the area in which they require support.

Expected Outcome

A completed skills and availability register for all 16 members.

Agenda 4: Final Selection of Project Domain

Estimated time: 10 minutes

The proposed project domain is:

Courier/Food Delivery Service

The proposed project title is:

Legon Smart Food Delivery and Rider Dispatch Optimizer

The system will represent a local food delivery service operating around areas such as:

- University of Ghana.
- Legon.
- East Legon.
- Madina.
- Haatso.
- Adenta.
- Nearby communities.

The project brief directly allows a courier or food delivery context involving vendors, customer zones, riders, order priority and route networks.

Members may raise brief technical concerns before voting.

Decision to Be Recorded

Final domain: Courier/Food Delivery Service_

Final project title: Smart Food Delivery

Expected Outcome

One project domain and title officially approved.

Agenda 5: Approval of the System Scope

Estimated time: 15 minutes

The group will approve the exact functions of the system.

Proposed System Operations

The system should be able to:

1. Store restaurants, customers, riders, orders and locations.
2. Add new food delivery requests.
3. Process normal orders using FIFO rules.
4. Prioritise urgent or delayed orders.
5. Assign available riders to orders.
6. Search for orders, riders, restaurants and locations.
7. Sort orders by time, urgency, distance or deadline.
8. Find the shortest route between locations.
9. Determine which locations are reachable.
10. Select orders under time, capacity or budget constraints.
11. Save and reload records from a database.
12. Record algorithm performance.
13. Export performance results.
14. Produce test evidence, tables and graphs.

The project must answer operational questions involving service priority, shortest paths, connectivity, optimisation and permanent data storage.

Proposed Main Data Entities

- Locations.
- Roads.
- Restaurants or vendors.
- Customers.
- Riders.
- Food orders or service requests.
- Resources.
- Algorithm runs.
- Audit events.

Expected Outcome

A clearly defined system boundary and list of approved operations.

Agenda 6: Approval of the Project Leadership Structure

Estimated time: 10 minutes

The following provisional leadership structure will be presented for approval.

Position	Proposed Member
Group Coordinator	Adom
Assistant Group Coordinator	Mubarack
Technical Lead and Lead Java Developer	Solomon
Database Lead	Enam KODJOH-KPAKPASSOU
Testing and Quality Assurance Lead	ma_wuli
Research and Problem Definition Lead	Leslie Normanyo
Documentation Lead	Ebenezer
Media and Video Lead	Kwesi Adom
Presentation Coordinator	Samuel
Attendance and Minutes Officer	Anthonia

Technical Leadership Arrangement

The Technical Lead will be responsible for:

- Java project setup.
- Custom data-structure implementation.
- Algorithm implementation.
- Object-oriented system design.
- Database connection support.
- Console interface.
- GitHub and code integration.
- Final technical troubleshooting.

Other members will contribute through:

- Database preparation.
- Dataset preparation.
- Testing.
- Trace tables.

- Pseudocode.
- Flowcharts.
- Performance experiments.
- Report writing.
- Video production.
- Presentation and oral-defence preparation.

Expected Outcome

All leadership and coordination positions officially assigned.

Agenda 7: Formation of Working Teams

Estimated time: 20 minutes

Members may belong to more than one team.

Team A: Technical Development and Integration

Proposed lead: Solomon

Responsibilities:

- Java development.
 - Custom data structures.
 - Algorithms.
 - Console interface.
 - GitHub repository.
 - Code integration.
 - Technical support for other teams.
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Team B: Database and Dataset

Proposed lead: Enam KODJOH-KPAKPASSOU

Proposed supporting members:

- Leslie Normanyo.
- Baaba.
- Ebenezer.
- Naa Atswei.
- 24kt-Armani.
- nana.
- Other interested members.

Responsibilities:

- Database schema.
- SQL scripts.
- CSV templates.
- Restaurants.
- Riders.
- Locations.
- Road connections.
- Orders.
- Data validation.
- Database evidence.

The minimum dataset includes 50 locations, 100 roads, 300 service requests, 30 resources and 30 algorithm-run records.

Team C: Testing, Correctness and Performance

Proposed lead: ma_wuli

Proposed supporting members:

- Richie.
- Prosper.
- Member currently displayed as “Isaac”.
- Kwesi Adom.
- Other interested members.

Responsibilities:

- Normal-case testing.
- Boundary-case testing.
- Invalid-input testing.
- Debugging.
- Trace tables.
- Correctness evidence.
- Runtime experiments.
- Performance-result recording.
- Graph creation.

The project requires at least 40 tests, six trace tables, correctness evidence and counterexamples.

Team D: Research, Documentation and Report

Proposed lead: Leslie Normanyo

Proposed supporting members:

- Leslie Normanyo.
- Baaba.
- Richie.
- ma_wuli.
- Prosper.
- Other interested members.

Responsibilities:

- Problem statement.
 - System assumptions.
 - System boundaries.
 - Local-context research.
 - Data dictionary.
 - System architecture.
 - Pseudocode.
 - Flowcharts.
 - Development log.
 - Screenshots.
 - Technical report.
 - Individual contribution records.
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Team E: Presentation, Oral Defence and Media

Proposed lead: Kwesi Adom for media activities.

Proposed supporting members:

- Prosper
- Samuel.

Responsibilities:

- Demonstration-video recording.
- Video editing.
- Presentation structure.
- Demonstration script.
- Oral-defence preparation.
- Practice sessions.
- Ensuring every member understands an assigned topic.

Expected Outcome

Every member belongs to at least one primary team and one support team.

Agenda 8: Assignment of Presentation and Oral-Defence Topics

Estimated time: 20 minutes

Every member must study and defend:

- One data structure.
- One algorithm or technical area.

This is required even where the actual Java implementation is completed by the Technical Lead.

Assignment Principle

Topics will be distributed in the following order:

1. Members who selected specific skills will receive related topics first.
2. Members willing to learn will receive suitable beginner or intermediate topics.
3. Members who did not complete the poll will receive topics from the remaining requirements.
4. Advanced topics will be supported with explanations, examples and practice sessions.
5. Topic exchanges will only be accepted when both members agree and the change is recorded.

Provisional Presentation Allocation

Member	Proposed Data Structure	Proposed Algorithm or Area
Solomon	Red-black tree	Dijkstra's algorithm
Mubarack	Binary Search Tree	Binary search
Enam KODJOH-KPAKPASSOU	B-tree/database index	Database searching and indexing
Kwesi Adom	Priority queue/heap	Greedy rider assignment
ma_wuli	Queue	Insertion sort
Prosper	Stack	Selection sort
Richie	Circular queue	Linear search

Leslie Normanyo	Linked list	Breadth-First Search
Baaba	Dynamic array	Merge sort
Ebenezer	Deque	Quicksort
Naa Atswei	Iterator	Search trace and correctness
24kt-Armani	Hash table	Collision handling and lookup
Member “Isaac”	Set and map	Testing and algorithm edge cases
nana	Disjoint set basics	Depth-First Search
Henry	Graph adjacency list	Dynamic programming
Member 16	Graph adjacency matrix	Prim and Kruskal algorithms

This table remains provisional until it is approved during the meeting.

What Every Member Must Understand

For each assigned topic, the member must be able to explain:

1. What the structure or algorithm does.
2. How it works.
3. Where it is used in the system.
4. Its input and output.
5. A simple example.
6. Its time complexity.
7. One advantage.
8. One limitation or edge case.
9. A trace table or diagram where required.

Expected Outcome

All 16 members receive presentation and oral-defence topics.

Agenda 9: Approval of Development Tools and Workflow

Estimated time: 10 minutes

The following tools are proposed:

Purpose	Proposed Tool
Programming language	Java
Database	SQLite or MySQL
Database connection	JDBC
Source-code management	GitHub
Documentation	Google Docs
File storage	Google Drive
Communication	WhatsApp
Task tracking	Google Sheets, Trello or GitHub Projects
Performance graphs	Excel, Python or Java
Video recording	OBS Studio or another agreed screen recorder

The brief recommends Java and permits SQLite, MySQL or PostgreSQL.

Workflow Rules to Approve

- One shared GitHub repository.
- One shared Google Drive folder.
- Clear file and folder naming.
- Tasks must have responsible members and deadlines.
- Completed tasks must include evidence.
- Code must not be pushed directly to the main branch without review.
- Difficulties must be reported early.
- Weekly progress must be recorded.
- Each member's contribution must be documented.

Expected Outcome

A single agreed development and collaboration process.

Agenda 10: Assignment of First Project Tasks

Estimated time: 15 minutes

The following tasks should be assigned before the meeting ends.

Task	Proposed Responsible Member(s)	Deadline
Final problem statement	Leslie and Documentation Team	
Final system scope	Technical and Research Teams	
Database schema	Enam and Database Team	
Data dictionary	Database and Documentation Teams	
Initial 50 locations	Dataset Team	
Initial 100 road connections	Dataset and Technical Teams	
Initial restaurant records	Dataset Team	
Initial rider records	Dataset Team	
First sample food orders	Dataset Team	
Java project structure	Solomon	
GitHub repository	Solomon	
Custom data-structure implementation plan	Solomon	
Search and sorting pseudocode	Mubarack	
Testing template	ma_wuli and Testing Team	
Development-log template	Documentation Team	
Flowchart and architecture draft	Mubarack and Documentation Team	
Demonstration-video plan	Henry and Kwesi	
Presentation-topic study notes	All members	

Expected Outcome

Every member leaves with at least one clear responsibility and deadline.

Agenda 11: Confirmation of Project Kick-Off

Estimated time: 5 minutes

The project will officially begin after the following have been approved:

- Project domain.
- Project title.
- System scope.
- Team structure.
- Leadership positions.
- Presentation topics.
- Development tools.
- First tasks.
- Deadlines.

Proposed Kick-Off Date

Agenda 12: Scheduling the Next Progress Meeting

Estimated time: 5 minutes

The next meeting must focus on reviewing completed work rather than repeating planning discussions.

Next meeting date: _____

Next meeting time: _____

Venue/platform: _____

5. Expected Commitment From Every Member

Every member is expected to:

1. Attend group meetings.
2. Complete assigned tasks before the deadline.
3. Report difficulties early.
4. Participate in at least one working team.
5. Study the assigned data structure and algorithm.
6. Prepare a simple explanation and example.
7. Contribute evidence such as data, tests, diagrams, notes or screenshots.
8. Attend presentation practice sessions.
9. Maintain respectful communication.

10. Avoid depending completely on the Technical Lead.
11. Understand the general purpose of the project.
12. Be prepared to answer questions during the oral defence.

The final submission includes source code, database scripts, performance results, a technical report, a demonstration video and an oral defence.

6. Expected Preparation for the Next Progress Meeting

After Meeting 2, every member must return to the next meeting with evidence of completed work.

Technical Lead

Expected to provide:

- Java project structure.
- GitHub repository.
- Initial model classes.
- First custom data structures.
- Proposed package structure.
- Database-connection plan.

Database Team

Expected to provide:

- Draft SQL schema.
- Table relationships.
- Sample records.
- CSV templates.
- Draft data dictionary.

Dataset Team

Expected to provide:

- List of locations.
- Road connections.
- Restaurant records.
- Rider records.

- Sample orders.
- Data-validation notes.

Testing Team

Expected to provide:

- Testing template.
- Initial test cases.
- Boundary cases.
- Invalid-input cases.
- Expected and actual result format.

Research and Documentation Team

Expected to provide:

- Problem statement.
- Project background.
- Ghanaian-context description.
- System assumptions.
- System boundaries.
- Initial report structure.
- Updated development log.

Media and Presentation Team

Expected to provide:

- Demonstration-video plan.
- Proposed presentation structure.
- Recording tools.
- Initial oral-defence preparation plan.

Every Individual Member

Expected to provide:

- A short explanation of the assigned data structure.
 - A short explanation of the assigned algorithm.
 - One example or diagram.
 - The use of the topic within the food delivery system.
 - A statement of progress on the assigned project task.
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7. Meeting 2 Expected Outcomes

Meeting 2 should not end until the group has:

- Recorded the details of all 16 members.
 - Finalised the project domain.
 - Approved the project title.
 - Defined the system scope.
 - Confirmed the leadership structure.
 - Assigned every member to working teams.
 - Assigned every member a data structure.
 - Assigned every member an algorithm or technical area.
 - Confirmed the database and development tools.
 - Created a task-allocation table.
 - Assigned deadlines.
 - Scheduled the next meeting.
 - Recorded attendance and meeting minutes.
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8. Meeting 2 Decision Record

Final Project Domain

Final Project Title

Final Database

Final Development Tools

Approved Group Coordinator

Approved Technical Lead

Approved Database Lead

Approved Testing Lead

Approved Documentation Lead

Approved Media and Presentation Lead

9. Action Register

Member	Primary Role	Support Role	Data Structure	Algorithm/Ar ea	First Task	Deadline
1. Solomon						
2. Mubarack						
3. Enam KODJOH-KPAKPASS OU						
4. Kwesi Adom						
5. ma_wuli						
6. Prosper						
7. Richie						
8. Leslie Normanyo						
9. Baaba						
10. Ebenezer						
11. Naa Atswei						
12. 24kt-Armani						

13. Member “Isaac”

14. nana

15.

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16.

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10. Meeting Closing Statement

Meeting 2 marks the official beginning of the project.

The Technical Lead will coordinate the main Java and algorithm implementation. However, the successful completion of the project will depend on the combined contribution of all 16 members through database work, dataset preparation, testing, trace tables, documentation, research, performance analysis, video production and oral-defence preparation.

Every member must leave the meeting with a clearly assigned responsibility, presentation topic and deadline.